

Phospho-mTOR expression in human glioblastoma microglia-macrophage cells.

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The glioblastoma (GBM) immune microenvironment is highly heterogeneous, and microglia may represent 30-70% of the entire tumor. However, the role of microglia and other specific immune populations is poorly characterized. Activation of mTOR signaling occurs in numerous human cancers and has roles in microglia-glioma cell interactions. We now show in human tumor specimens (42 patients), that 39% of tumor-associated microglial (TAM) cells express mTOR phosphorylated at Ser-2448; and similar mTOR activation is observed using a human microglia-glioma interaction paradigm. In addition, we confirm previous studies that microglia express urea and ARG1 (taken as M2 marker) in the presence of glioma cells, and this phenotype is down-regulated in the presence of a mTOR inhibitor. These results suggest that mTOR suppression in GBM patients might induce a reduction of the M2 phenotype expression in up to 40% of all TAMs. Since the M2 profile of microglial activation is believed to be associated with tumor progression, reductions in that phenotype may represent an additional anti-tumor mechanism of action of mTOR inhibitors, along with direct anti-proliferative activities.

Glioblastoma cells: a heterogeneous and fatal tumor interacting with the parenchyma.

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Glioblastomas (GBMs) are considered to be one of the deadliest human cancers, characterized by a high proliferative rate, aggressive invasiveness and insensitivity to radio- and chemotherapy, as well as a short patient survival period. Moreover, GBMs are among the most vascularized and invasive cancers in humans. Angiogenesis in GBMs is correlated with the grade of malignancy and is inversely correlated with patient survival. One of the first steps in tumor invasions is migration. GBM cells have the ability to infiltrate and disrupt physical barriers such as basement membranes, extracellular matrix and cell junctions. The invasion process includes the overexpression of several members of a super-family of zinc-based proteinases, the Metzincin, in particular a sub-group, metalloproteinases. Another interesting aspect is that, inside the GBM tissue, there are up to 30% of microglia or macrophages. However, little is known about the immune performance and interactions of the microglia with GBMs. These singular properties of GBMs will be described here. A sub-population of cells with stem-like properties may be the source of tumors since, apparently, GBM stem cells (GSCs) are highly resistant to current cancer treatments. These cancer therapies, while killing the majority of tumor cells, ultimately fail in GBM treatment because they do not eliminate GSCs, which survive to regenerate new tumors. Finally, GBM patient prognostic has shown little improvement in decades. In this context, we will discuss how the membrane-acting toxins called cytolytins can be a potential new tool for GBM treatment.

MCP-1 and MIP-3 α Secreted from Necrotic Cell-Treated Glioblastoma Cells Promote Migration/Infiltration of Microglia.

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Glioblastoma multiforme (GBM) is the most common primary brain tumor in adults. The defining characteristics of GBM are diffuse infiltration of tumor cells into normal brain parenchyma, rapid growth, a high degree of infiltration of microglia and macrophages, and the presence of necrosis.

Microglia/macrophages are frequently found in gliomas and they extensively infiltrate GBM tissue, up to 30% of total tumor mass. However, little is known about the effect of necrotic cells (NCs) on microglia infiltration in GBM and the tumor-infiltrating microglia-induced factors in GBMs. In this study, to address whether necrosis or necrosis-exposed GBM cells affect the degree of microglia/macrophage infiltration, migration and invasion/infiltration assays were performed. Culture supernatants and nuclear extracts of CRT-MG cells treated or untreated with necrotic cells were analyzed using a chemokine array and electrophoretic mobility shift assay, respectively. The presence of NCs promoted the migration/infiltration of microglia, and GBM cell line CRT-MG cells exposed to NCs further enhanced the migration and infiltration of HMO6 microglial cells. Treatment with NCs induced mRNA and protein expression of chemokines such as Monocyte Chemoattractant Protein-1 (CCL2/MCP-1) and Macrophage Inflammatory Protein-3 α (CCL20/MIP-3 α) in CRT-MG cells. In particular, CCL2/MCP-1 and CCL20/MIP-3 α were significantly increased in NC-treated CRT-MG cells. NCs induced DNA binding of the transcription factors Nuclear Factor (NF)- κ B and Activator Protein 1 (AP-1) to the CCL2/MCP-1 and CCL20/MIP-3 α promoters, leading to increased CCL2/MCP-1 and CCL20/MIP-3 α mRNA and protein expression in CRT-MG cells. These results provide evidence that NCs induce the expression of CCL2/MCP-1 and CCL20/MIP-3 α in glioblastoma cells through activation of NF- κ B and AP-1 and facilitate the infiltration of microglia into tumor tissues.

Microglia-glioma cross-talk: a two way approach to new strategies against glioma.

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Glioblastoma (GBM) is the most malignant and aggressive among primary brain tumors, characterized by very low life expectancy. In vivo, glioma and glioblastoma in particular contain large numbers of immune cells (myeloid cells) such as microglia and tumour-infiltrating macrophages (or glioma associated macrophages). These glioma-infiltrating myeloid cells comprise up to 30% of total tumor mass and have been suggested to play several roles in glioma progression including proliferation, survival, motility and immunosuppression. Although tumor microglia and macrophages can acquire proinflammatory (M1) phenotype being capable of releasing proinflammatory cytokines, phagocytosing and presenting antigens, their effector immune function in gliomas appears to be suppressed by the acquisition of an anti-inflammatory (M2) phenotype. In the present work we review the microglia-glioma interactions to highlight the close relationship between the two cell types and the factors that can influence their properties (chemokines, cytokines, S100B protein). A future therapeutic possibility might be to simultaneously targeting, for example with nanomedicine, glioma cells and microglia to push the microglia towards an antitumor phenotype (M1) and/or prevent glioma cells from "conditioning" by microglia.

Exploring Monocytes-Macrophages in Immune Microenvironment of Glioblastoma for the Design of Novel Therapeutic Strategies.

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Gliomas are primary malignant brain tumors. These tumors seem to be more and more frequent, not only because of a true increase in their incidence, but also due to the increase in life expectancy of the general population. Among gliomas, malignant gliomas and more specifically glioblastomas (GBM) are a

challenge in their diagnosis and treatment. There are few effective therapies for these tumors, and patients with GBM fare poorly, even after aggressive surgery, chemotherapy, and radiation. Over the last decade, it is now appreciated that these tumors are composed of numerous distinct tumoral and non-tumoral cell populations, which could each influence the overall tumor biology and response to therapies. Monocytes have been proved to actively participate in tumor growth, giving rise to the support of tumor-associated macrophages (TAMs). In GBM, TAMs represent up to one half of the tumor mass cells, including both infiltrating macrophages and resident brain microglia. Infiltrating macrophages/monocytes constituted ~ 85% of the total TAM population, they have immune functions, and they can release a wide array of growth factors and cytokines in response to those factors produced by tumor and non-tumor cells from the tumor microenvironment (TME). A brief review of the literature shows that this cell population has been increasingly studied in GBM TME to understand its role in tumor progression and therapeutic resistance. Through the knowledge of its biology and protumoral function, the development of therapeutic strategies that employ their recruitment as well as the modulation of their immunological phenotype, and even the eradication of the cell population, can be harnessed for therapeutic benefit. This revision aims to summarize GBM TME and localization in tumor niches with special focus on TAM population, its origin and functions in tumor progression and resistance to conventional and experimental GBM treatments. Moreover, recent advances on the development of TAM cell targeting and new cellular therapeutic strategies based on monocyte/macrophages recruitment to eradicate GBM are discussed as complementary therapeutics.

Single-cell image analysis reveals a protective role for microglia in glioblastoma.

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Microglia and tumor-associated macrophages (TAMs) constitute up to half of the total tumor mass of glioblastomas. Despite these myeloid populations being ontogenetically distinct, they have been largely conflated. Recent single-cell transcriptomic studies have identified genes that distinguish microglia from TAMs. Here we investigated whether the translated proteins of genes enriched in microglial or TAM populations can be used to differentiate these myeloid cells in immunohistochemically stained human glioblastoma tissue. Tissue sections from resected low-grade, meningioma, and glioblastoma (grade IV) tumors and epilepsy tissues were immunofluorescently triple-labeled for Iba1 (pan-myeloid marker), CD14 or CD163 (preferential TAM markers), and either P2RY12 or TMEM119 (microglial-specific markers). Using a single-cell-based image analysis pipeline, we quantified the abundance of each marker within single myeloid cells, allowing the identification and analysis of myeloid populations. P2RY12 and TMEM119 successfully discriminated microglia from TAMs in glioblastoma. In contrast, CD14 and CD163 expression were not restricted to invading TAMs and were upregulated by tumor microglia. Notably, a higher ratio of microglia to TAMs significantly correlated with increased patient survival. We demonstrate the validity of previously defined microglial-specific genes P2RY12 and TMEM119 as robust discriminators of microglia and TAMs at the protein level in human tissue. Moreover, our data suggest that a higher proportion of microglia may be beneficial for patient survival in glioblastoma. Accordingly, this tissue-based method for myeloid population differentiation could serve as a useful prognostic tool.

The prognostic value of tumor-associated macrophages in glioma patients.

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Glioma is a complex tumor composed of both neoplastic and non-neoplastic cells, including tumor-infiltrating leukocytes (TILs), and each cell type contributes to tumor formation and malignant progression. Among TILs, tumor-associated macrophages (TAMs) are of great importance and play a key role in the immune response to cancer. In this study, 22 types of adaptive and innate TILs were evaluated in gliomas. TAMs, which account for 38.7% of all these cells, are the most abundant immune infiltrates in the tumor microenvironment. In addition, we observed different immune cell patterns in low-grade glioma and glioblastoma. Our research indicated that there was a connection between TILs, and 13 of 22 TILs were significantly associated with patient outcomes. Finally, the prognosis and diagnostic value of TAMs were revealed using Kaplan-Meier analysis. We identified the optimal cutoff point of TAMs at an infiltrating level of 0.47 to predict patient prognosis, with a median overall survival of 448 days in patients with higher TAM infiltration levels and 2660 days in patients with lower TAM infiltration levels. These findings provide a new idea for glioma to regulate tumor-specific immunity, clarify the potential effects of TAMs on disease pathology, and provide a theoretical basis for immune intervention treatment of gliomas.

Oncolytic Virotherapy Blockade by Microglia and Macrophages Requires STAT1/3.

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The first oncolytic virotherapy employing HSV-1 (oHSV-1) was approved recently by the FDA to treat cancer, but further improvements in efficacy are needed to eradicate challenging refractory tumors, such as glioblastomas (GBM). Microglia/macrophages comprising approximately 40% of a GBM tumor may limit virotherapeutic efficacy. Here, we show these cells suppress oHSV-1 growth in gliomas by internalizing the virus through phagocytosis. Internalized virus remained capable of expressing reporter genes while viral replication was blocked. Macrophage/microglia formed a nonpermissive OV barrier, preventing dissemination of oHSV-1 in the glioma mass. The deficiency in viral replication in microglial cells was associated with silencing of particular viral genes. Phosphorylation of STAT1/3 was determined to be responsible for suppressing oHSV-1 replication in macrophages/microglia. Treatment with the oxindole/imidazole derivative C16 rescued oHSV-1 replication in microglia/macrophages by inhibiting STAT1/3 activity. In the U87 xenograft model of GBM, C16 treatment overcame the microglia/macrophage barrier, thereby facilitating tumor regression without causing a spread of the virus to normal organs. Collectively, our results suggest a strategy to relieve a STAT1/3-dependent therapeutic barrier and enhance oHSV-1 oncolytic activity in GBM. Significance: These findings suggest a strategy to enhance the therapeutic efficacy of oncolytic virotherapy in glioblastoma. Cancer Res; 78(3); 718-30. ©2017 AACR.

The brain tumor microenvironment.

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High-grade brain tumors are heterogeneous with respect to the composition of bona fide tumor cells and with respect to a range of intermingling parenchymal cells. Glioblastomas harbor multiple cell types, some with increased tumorigenicity and stem cell-like capacity. The stem-like cells may be the cells of origin for tumor relapse. However, the tumor-associated parenchymal cells—such as vascular cells, microglia, peripheral immune cells, and neural precursor cells—also play a vital role in controlling the course of pathology. In this review, we describe the multiple interactions of bulk glioma cells and glioma

stem cells with parenchymal cell populations and highlight the pathological impact and signaling pathways known for these types of cell-cell communication. The tumor-vasculature not only nourishes glioblastomas, but also provides a specialized niche for these stem-like cells. In addition, microglial cells, which can contribute up to 30% of a brain tumor mass, play a role in glioblastoma cell invasion. Moreover, non-neoplastic astrocytes can be converted into a reactive phenotype by the glioma microenvironment and can then secrete a number of factors which influences tumor biology. The young brain may have the capacity to inhibit gliomagenesis by the endogenous neural stem and progenitor cells, which secrete tumor suppressive factors. The factors, pathways, and interactions described in this review provide a new prospective on the cell biology of primary brain tumors, which may ultimately generate new treatment modalities. However, our picture of the multiple interactions between parenchymal and tumor cells is still incomplete. © 2011 Wiley-Liss, Inc.

In Vitro Assay to Study Tumor-macrophage Interaction.

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Tumor associated macrophages (TAMs) account for a large percentage of cells in the tumor mass for different types of cancers. Glioblastoma (GBM), a malignant brain tumor with no cure, has up to a half the tumor mass TAMs. TAMs can be pro-tumoral or anti-tumoral, depending on the activation of specific genes in the cells. Genetic mutations in the tumors, through regulating cytokine expression, can affect recruitment of TAMs to the tumor microenvironment. Here, we describe a quantitative cell-based assay to assess macrophage recruitment by the conditioned medium from the tumor cells. This assay uses the human macrophage cell line MV-4-11 to study macrophage attraction by the conditioned medium from glioblastoma, allowing for high reproducibility and low variability. Data generated with this assay can contribute to a better understanding of the interaction between the tumor and the tumor microenvironment. Similar assay can be used to assess interaction between the tumor cells and other immune cells, including T cells and natural killer (NK) cells.
